

PROFILE

Jinmin Kim

Robot Software Engineer - 3D Registration, Medical Imaging, Robot Systems

An R&D engineer whose work spans surgical navigation, XR, 3D registration, multi-sensor integration, and autonomous-forklift field deployment. I connect coordinate models and algorithms to physical devices, user interfaces, and explicit validation flows.

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EDUCATION

DGIST

2021.03 - 2023.02

M.S. in Robotics and Mechatronics Engineering

- Surgical Robotics and Augmented Reality Laboratory (advisor: Jaesung Hong) - surgical navigation team
- Thesis: Virtual Jawbone Reduction Planning based on Dental Occlusion for Mandibular Fracture Surgery
- Multi-centre oral and maxillofacial AR surgery programme (2021.06 - 2023.05) - early and supporting development of the Unity-based AR navigation

Kumoh National Institute of Technology

2015.03 - 2021.02

B.S. in Mechanical System Engineering

- Undergraduate researcher, System & Vision Lab. (2019 - 2020); completed the DGIST and UNIST summer undergraduate research programmes
- Designed and built a five-bar-linkage quadruped robot (2020.04 - 2021.01) - mechanism design, kinematic modelling, and control, leading to a patent application and an engineering-fair award
- Vice president of the invention club (2019) - several patent applications and invention-contest awards; the club was designated an outstanding education-donation club by the Ministry of Education

EXPERIENCE

DIGITRACK Inc.

2023.02 - Present

Researcher · Software Development

A DGIST spin-off medical-device and robotics company (marker-based 3D optical tracking devices and software), Daegu

Surgical navigation and medical imaging

- Over three years of collaboration with the oral and maxillofacial surgery department of Samsung Medical Center - biweekly clinical meetings since 2023, with two government programmes and one development contract carried as the main engineer.
- Sole developer of the 3D Slicer-based surgical-navigation software in use at the hospital - registering and overlaying the surgical plan through optical tracking, and building an AR surgical guide on a registered HoloLens 2 coordinate frame.
- Built a patient-and-clinician consultation XR application on Meta Quest, synchronising spatial, voice, and scenario state across users.
- Led development of the NeuroPilot rTMS coil-navigation system - digital-twin visualisation of coil pose (health-technology government programme).
- Low-dose C-arm 3D navigation - connected C-arm imaging to the 3D navigation workflow and implemented the owned feature scope.
- Own surface-guided respiratory tracking and 4D CBCT registration for radiotherapy (large KERI programme, 2026.06 -).
- Delivered several development contracts for research institutes including KERI and KIMM.

EXPERIENCE

3D optical tracking software

- Develop and maintain the API and Viewer software for SKADI, the in-house optical 3D tracker - the software layer of a product line delivered to Korean surgical-navigation companies and research institutes.
- Wrote a 3D Slicer custom-application template so research groups can start a navigation prototype immediately.

Industrial robotics - autonomous forklift (2025 -)

- Designed and implemented the safety-policy manager for the in-house autonomous-forklift solution (DOTORI) and integrated LiDAR and ToF sensors over the Zenoh and ROS 2 network.
- Own vision R&D - truck-loading vision with ToF and RGB fusion, a SAM-family segmentation proof of concept, and behaviour-tree integration.
- Deployed, commissioned, and operated the system at two production sites, including the field-deployment engineering.

Personal - building with AI (2026 -)

- LLM Wiki - built and operate a personal knowledge system driven by AI agents. I own the framing, architecture, and verification while delegating much of the implementation to agents, and run the real-data migration and daily automation routines (TypeScript, Python, PostgreSQL, MCP).

PUBLICATIONS AND PRESENTATIONS

- 2026** Development and Clinical Evaluation of an Extended Reality-Based Consultation Platform "OMFS VR Consultation App" for Enhanced Patient Understanding in Orthognathic Surgery
2026 Spring Conference, Korean Society of Medical Metaverse · Poster presentation · Co-author
- 2024** A Proof of Concept: Optimized Jawbone-Reduction Model for Mandibular Fracture Surgery
Journal of Imaging Informatics in Medicine (SCIE, Q1) · Journal article · Joint first author
[Public article](#)
- 2024** Set-up of Dental Final Occlusion using Haptic Device in Orthognathic Surgery
2024 Spring Conference, Korean Society of Medical Metaverse · Poster presentation · Co-author · Best Poster Award
- 2023** Occlusion-based Jawbone Model Optimization for Mandibular Fracture Surgery
14th Conference, Korean Society of Medical Robotics · Oral presentation · Co-author · Best Paper Award
- 2022** Dental Occlusion Model Using Arch Line for Mandibular Fracture Surgery
ACCAS 2022, Bangkok, Thailand · International conference paper · First author
- 2022** Virtual Jawbone Model Reconstruction for Mandibular Fracture Surgery Guidance
13th Conference, Korean Society of Medical Robotics · Oral presentation · First author
- 2019** Design of Ping-Pong Ball Launcher
ISM 2019 - International Symposium on Mechatronics · Conference paper · First author

PATENTS · 7 APPLICATIONS / 3 GRANTED

- Granted** Coordinate-frame registration method for real-time 3D tracking of surgical instruments
10-2024-0186869 · 2024.12 filed · Employment
- Granted** Position tracking device and method
10-2024-0186864 · 2024.12 filed · Employment
- Filed** Two-axis rotating stereo-camera actuation device and method
10-2024-0127937 · 2024.09 filed · Employment
- Filed** Quadruped walking robot using a multi-joint link structure
10-2021-0008332 · 2021.01 filed · Undergraduate
- Granted** Clothes hanger with an elastic section
10-2019-0100328 · 2019.08 filed · Undergraduate
- Filed** Assemblable carrier with adjustable storage space
10-2019-0010185 · 2019.01 filed · Undergraduate
- Filed** Disposable paper-cup collection bin
10-2015-0122661 · 2015.08 filed · Undergraduate

HONOURS AND AWARDS · 9 TOTAL

- 2024** Best Poster Award - Korean Society of Medical Metaverse
- 2023** Best Paper Award - Korean Society of Medical Robotics
- 2020** Encouragement Prize, Engineering Fair (five-bar-linkage quadruped robot) - Kumoh National Institute of Technology
- 2020** Bronze Prize, ROS-based autonomous-driving training - Training programme competition
- 2019** Grand Prize, International TRIZ Competition - The International TRIZ Association
- 2019** Grand Prize, Startup Idea Competition - Kumoh National Institute of Technology
- 2019** Sponsor's Award, University Creative Invention Contest - Korea Invention Promotion Association
- 2019** Excellence Award, GREEN Earth Competition - Hyosung
- 2015** Excellence Award, University Creative Invention Contest - Korea Invention Promotion Association

SKILLS

- Programming** Python, C++, C#
- Medical imaging** 3D Slicer, DICOM, image registration (rigid, ICP), segmentation (SAM family)
- 3D and XR** Unity, HoloLens 2 (MRTK), Meta Quest, optical tracking
- Robotics** ROS 2, Zenoh, Isaac Sim, behaviour trees
- AI tooling** PyTorch, LLM-agent-based development workflow
- Languages** English - Intermediate (paper writing and reading) · Japanese - JLPT N2 (2021)

A CV built only from public facts approved on 2026-08-22. Personal information about other people and unverified outcome claims are excluded.